

# TARGET-90

**JKBOSE CLASS-11<sup>TH</sup>**

## Rotational Motion

One Shot

**PHYSICS**

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## Topics to be Covered

**A** Centre of Mass

**B** Why We Use COM

**C** Centre of Mass of Two Bodies System

**D** Motion of COM (*velocity*)

**E** Acceleration of COM





## Topics to be Covered

**F** Rotational Motion

**G** Angular Distance/Angular Velocity/Angular Acceleration

**H** Angular Momentum

**I** Torque

**J** Moment Of Inertia



## Topics to be Covered

**K** Radius of Gyration

**L** Law of Conservation of Momentum (Most important)

**M** ..

**N** ..



## System of Particles and Rotational Motion

$\hbar/\omega$   
 $m-$

Centre of mass of a two-particle system, momentum conservation and Centre of mass motion. Centre of mass of a rigid body; centre of mass of a uniform rod. Moment of a force, torque, angular momentum, law of conservation of angular momentum and its applications. Equilibrium of rigid bodies, rigid body rotation and equations of rotational motion, comparison of linear and rotational motions. Moment of inertia, radius of gyration, values of moments of inertia for simple geometrical objects (no derivation)



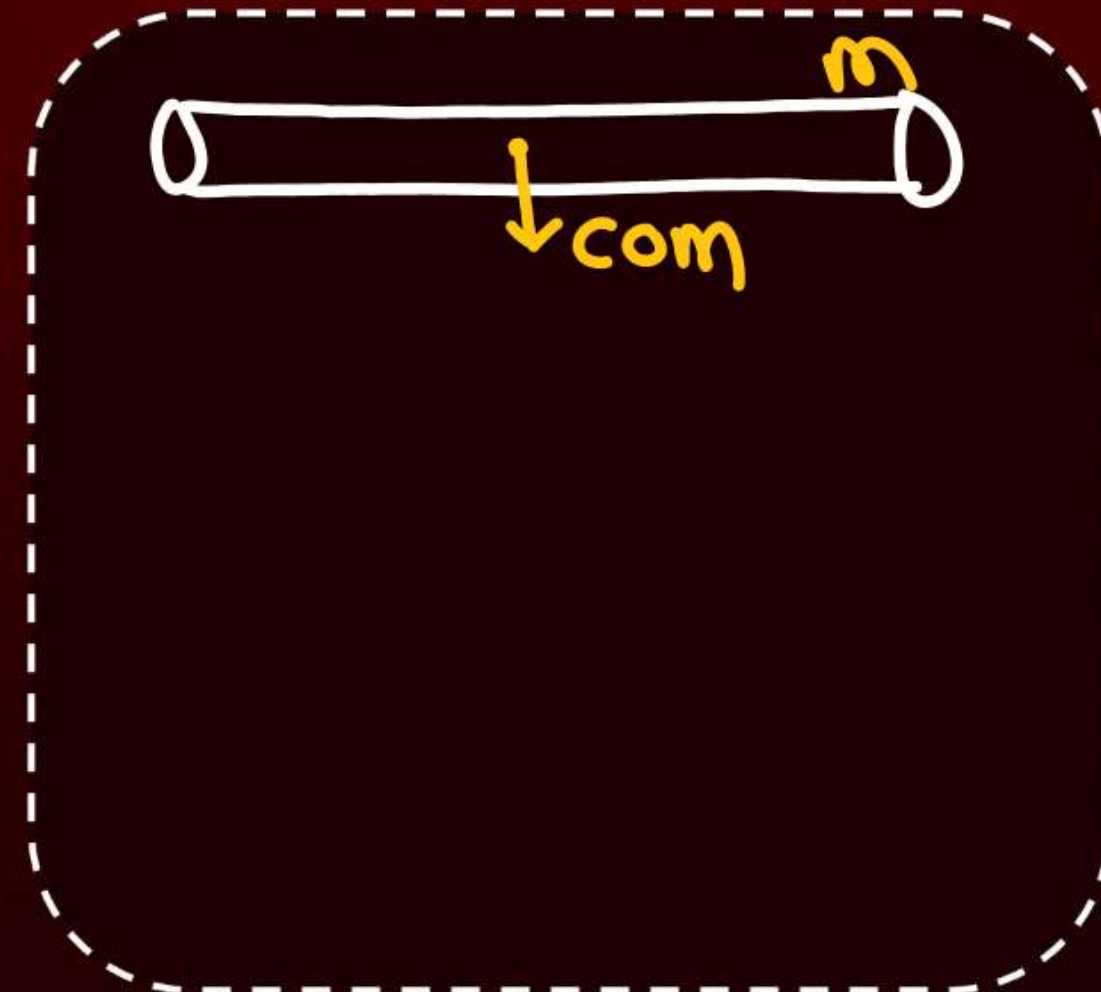




## Centre Of Mass

- ★ COM of masss is basically that point where the entire MASS of system is assumed to be CONCENTRATED

### \* Balancing point

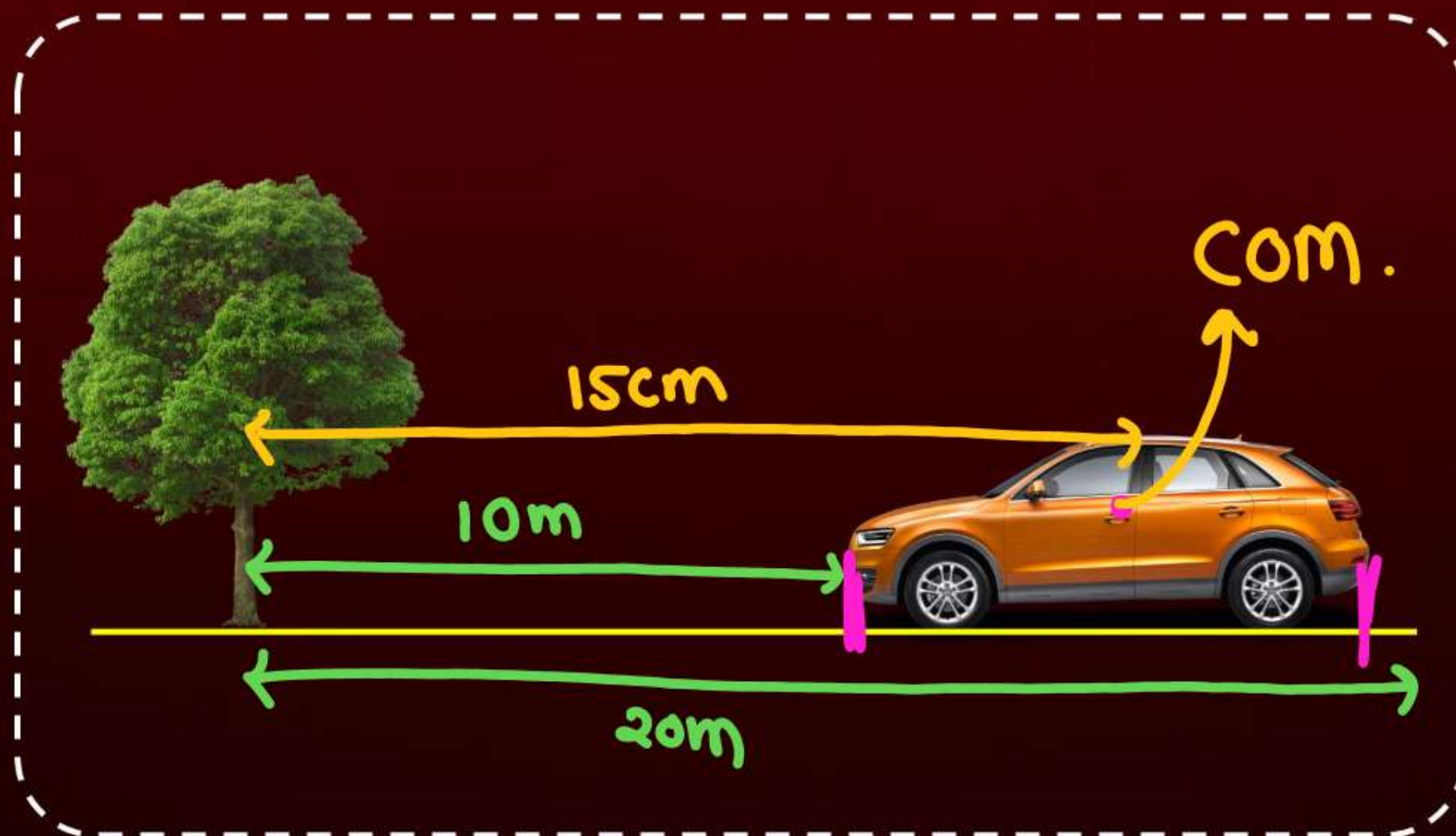






## Why We Use COM

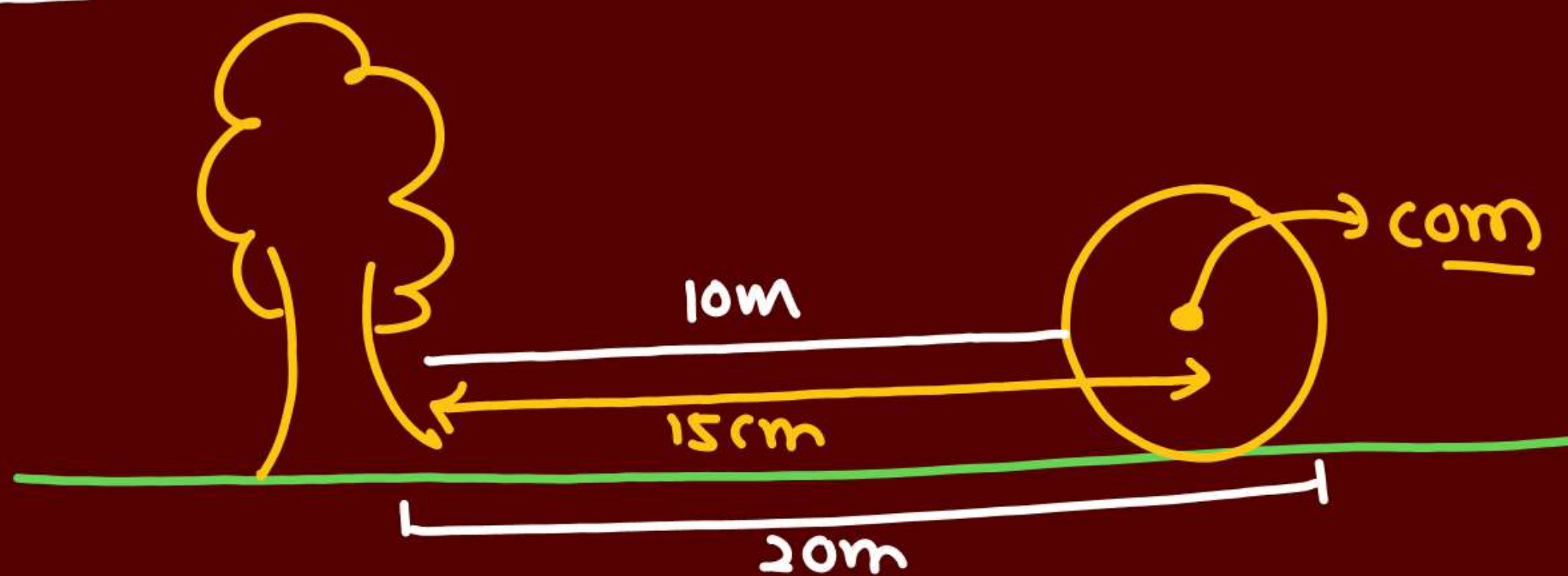
- ✦ It makes it easy to solve mechanics problems where we have to describe the oddly-shaped objects and complicated systems.



Note

Some time the com is located at that point where there is  
no mass

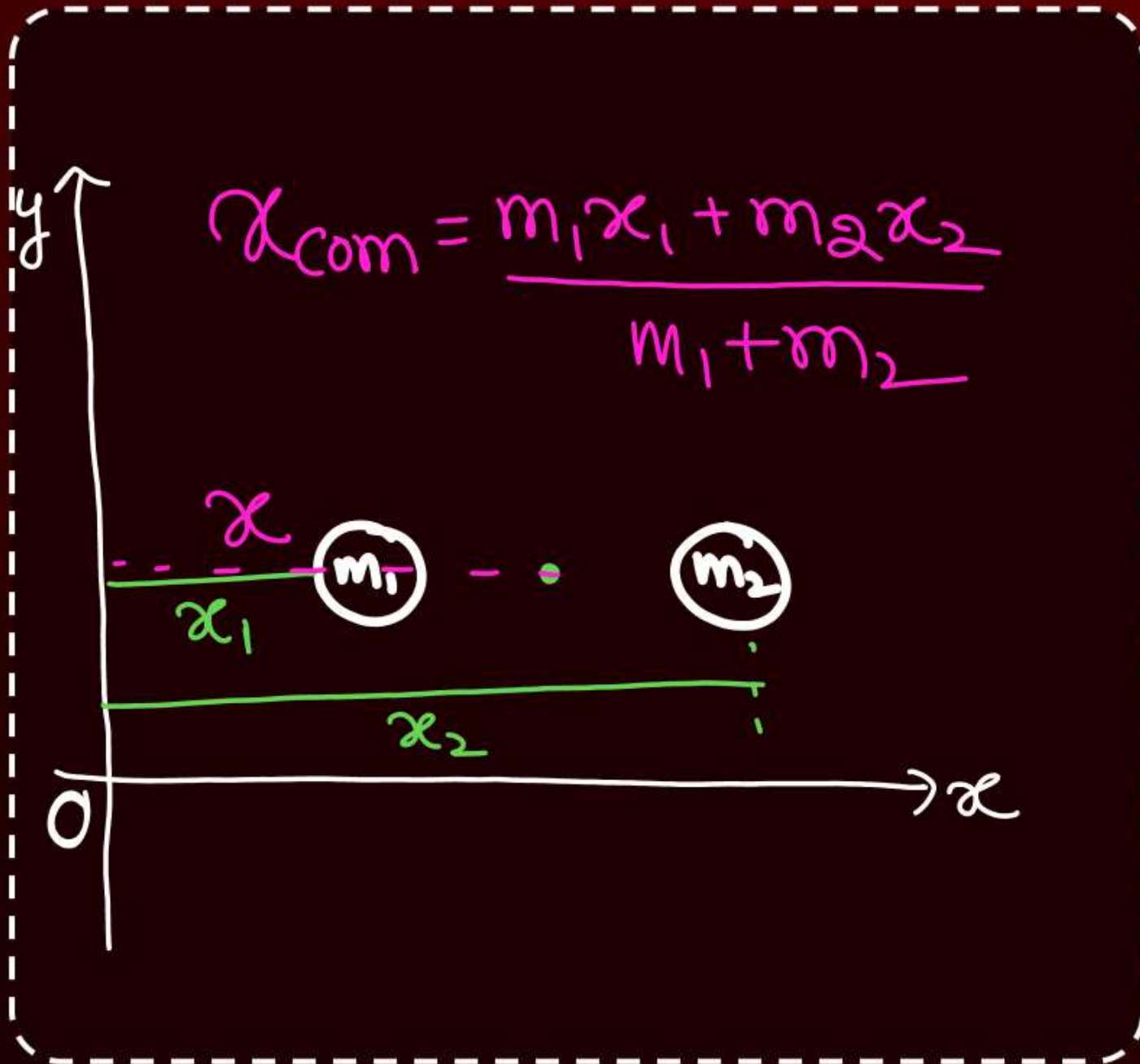
MLQ  
2023







## COM Of Mass Two Boder

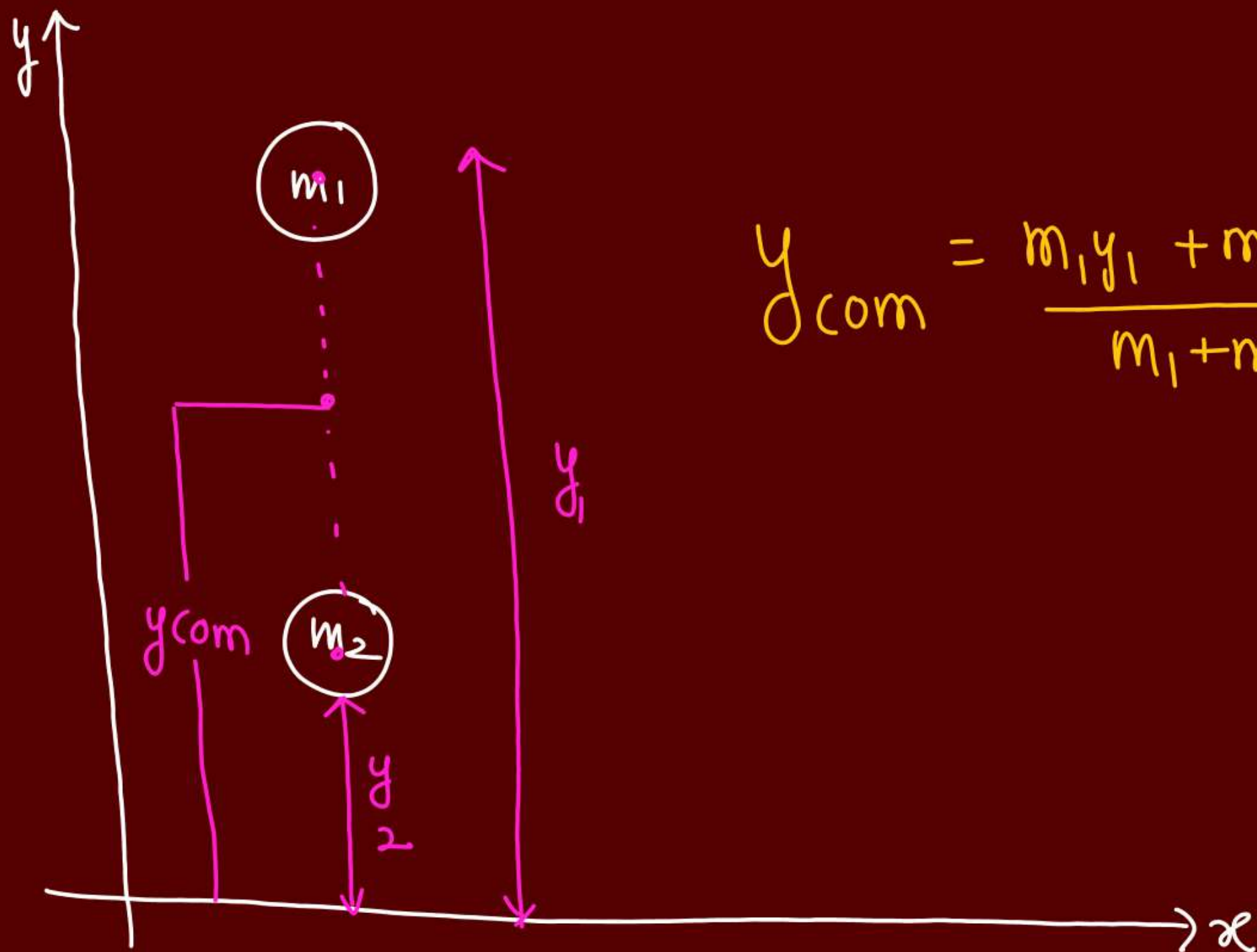


$$x_{com} = \frac{m_1 \vec{x}_1 + m_2 \vec{x}_2}{m_1 + m_2}$$

$$y_{com} = \frac{m_1 \vec{y}_1 + m_2 \vec{y}_2}{m_1 + m_2}$$

$$z_{com} = \frac{m_1 \vec{z}_1 + m_2 \vec{z}_2}{m_1 + m_2}$$

$$\vec{x}_{com} = \frac{m_1 \vec{x}_1 + m_2 \vec{x}_2}{m_1 + m_2}$$



$$y_{com} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$





## Motion Of COM

# The com moves It has velocity k/w velocity of com

# Velocity is Rate of change of position

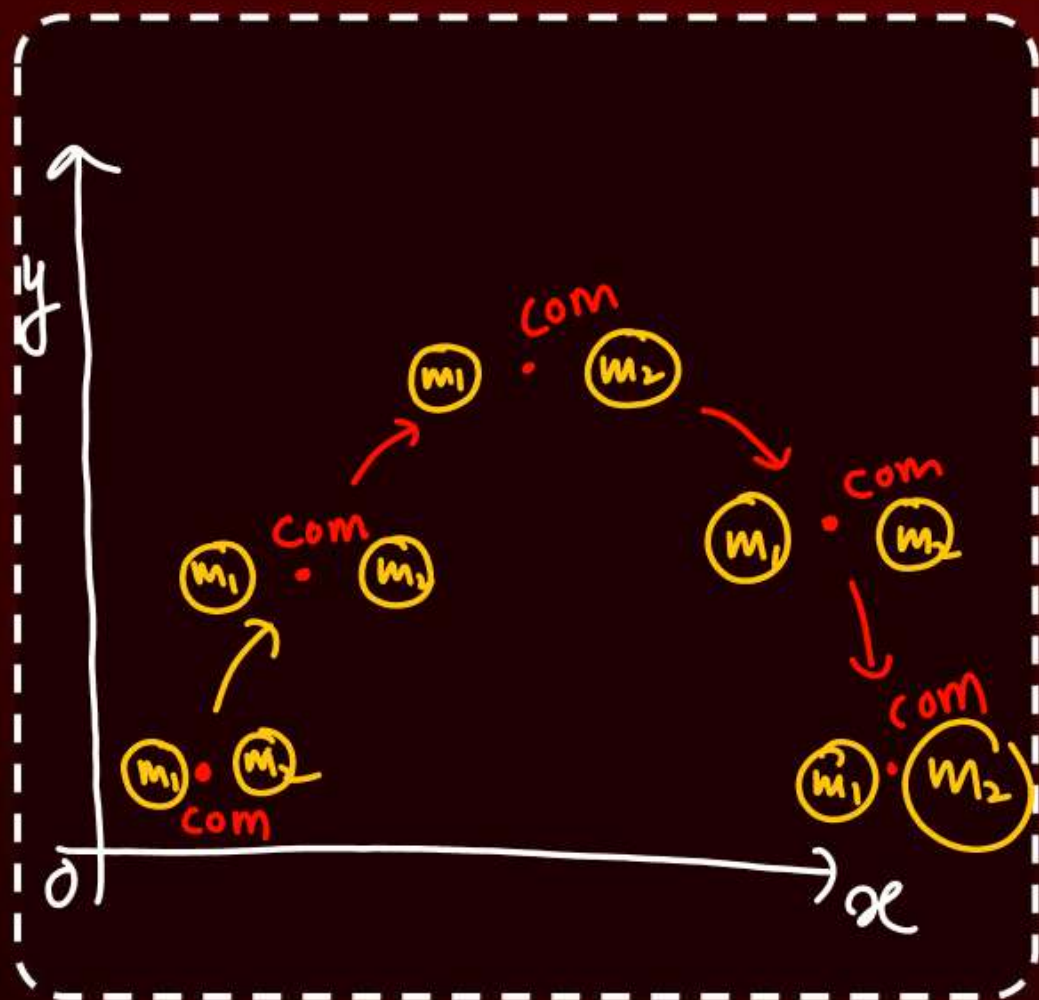
$$\# \vec{V}_{com} = \frac{d\vec{r}_{com}}{dt} \quad \vec{r}_{com} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

$$\vec{V}_{com} = \frac{d}{dt} \left( \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2} \right)$$

$$\vec{V}_{com} = \frac{1}{m_1 + m_2} \frac{d(m_1\vec{r}_1 + m_2\vec{r}_2)}{dt}$$

$$\vec{V}_{com} = \frac{1}{m_1 + m_2} \left( \frac{dm_1\vec{r}_1}{dt} + \frac{dm_2\vec{r}_2}{dt} \right)$$

$$\vec{V}_{com} = \frac{1}{m_1 + m_2} \left( m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} \right)$$



$$\vec{V}_{com} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

$$\vec{v}_{com} = \frac{1}{(m_1 + m_2)} \left( m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} \right)$$

$$\vec{v}_{com} = \frac{1}{m_1 + m_2} \left( m_1 \vec{v}_1 + m_2 \vec{v}_2 \right)$$

$$\vec{v}_{com} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$$

$$\vec{v}_{xcom} = \frac{m_1 \vec{v}_{x1} + m_2 \vec{v}_{x2}}{m_1 + m_2}$$

$$\vec{v}_{ycom} = \frac{m_1 \vec{v}_{y1} + m_2 \vec{v}_{y2}}{m_1 + m_2}$$

$$\vec{v}_{zcom} = \frac{m_1 \vec{v}_{z1} + m_2 \vec{v}_{z2}}{m_1 + m_2}$$



## Acceleration of com

# The rate change of  $\vec{v}_{com}$  is the acceleration of com

$$\vec{a}_{com} = \frac{d\vec{v}_{com}}{dt}$$

$$\vec{v}_{com} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

$$\vec{v}_{com}(m_1 + m_2) = m_1\vec{v}_1 + m_2\vec{v}_2$$

on differentiating b/s

$$\frac{d}{dt}(m_1 + m_2)\vec{v}_{com} = \frac{d}{dt}(m_1\vec{v}_1 + m_2\vec{v}_2)$$

$$\frac{d}{dt}(m_1 + m_2)\vec{v}_{com} = \frac{d}{dt}m_1\vec{v}_1 + \frac{d}{dt}m_2\vec{v}_2$$

$$a_{com}(m_1 + m_2) = m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt}$$

$$\vec{a}_{com}(m_1 + m_2) = m_1\vec{a}_1 + m_2\vec{a}_2$$

$$\vec{a}_{com} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2}$$





## Motion Of COM

$$\vec{a}_{com} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{m_1 + m_2}$$

$$\vec{a}_{xcom} = \frac{m_1 \vec{a}_{x1} + m_2 \vec{a}_{x2}}{m_1 + m_2}$$

$$\vec{a}_{ycom} = \frac{m_1 \vec{a}_{y1} + m_2 \vec{a}_{y2}}{m_1 + m_2}$$



$$\vec{a}_{zcom} = \frac{m_1 \vec{a}_{z1} + m_2 \vec{a}_{z2}}{m_1 + m_2}$$

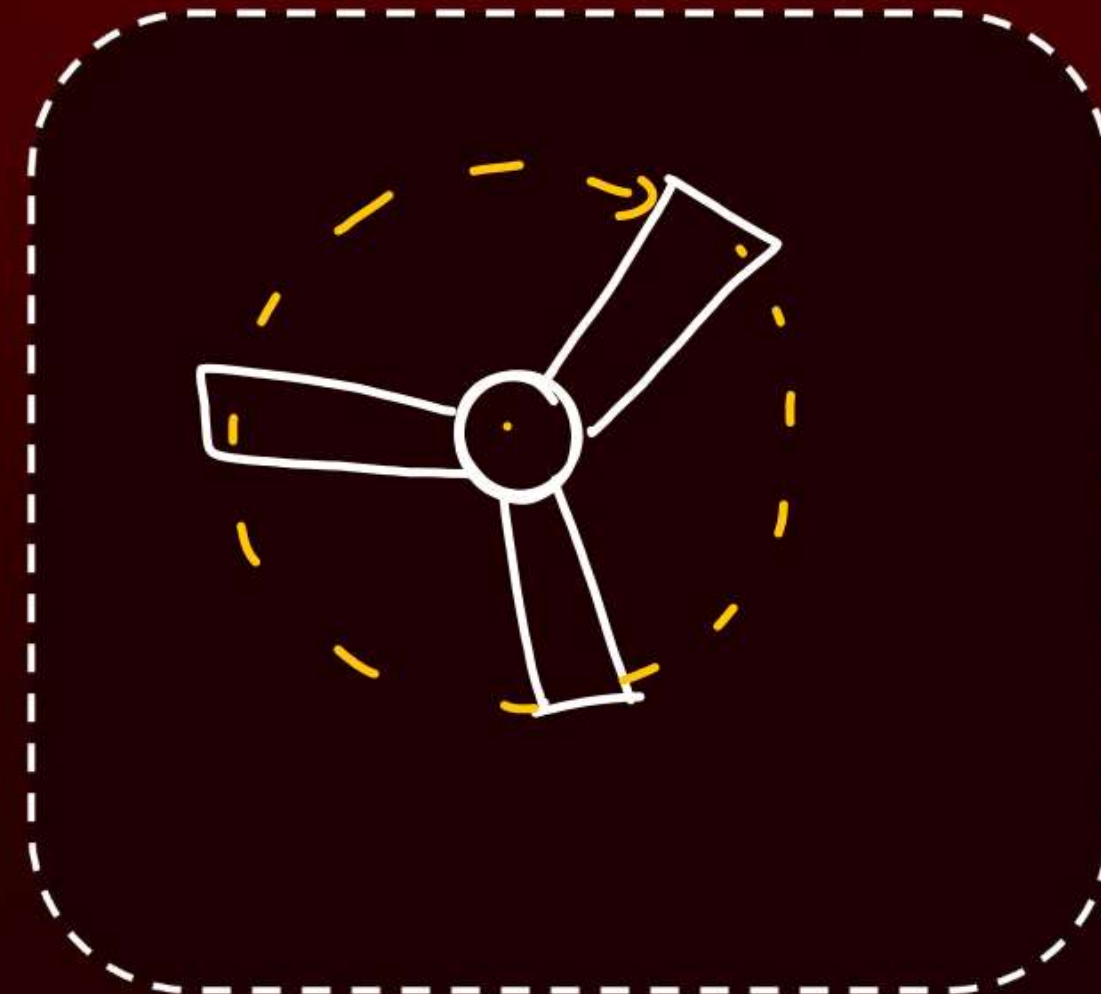




## Rotational Motion

- ✦ Rotational motion can be defined as the motion of an object around a circular path about fixed line (line of Axis)

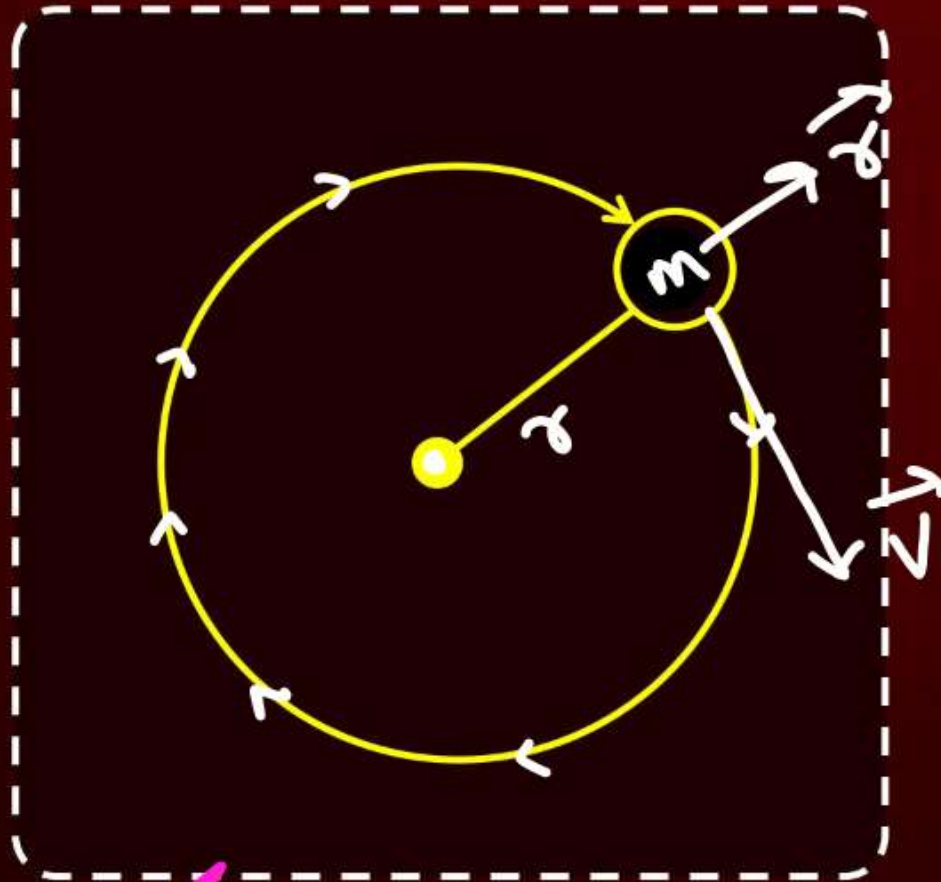
Axis of Rotation







# Angular Momentum ( $\vec{L}$ )



$$p = mv$$

✦ The momentum of the body when the body is rotated is called Angular Momentum. ✓

✦ It is the cross product of  $\vec{r}$  and  $\vec{p}$

$$\# \vec{L} = \vec{r} \times \vec{p}$$

$$\# \vec{L} = rp \sin \theta \hat{n}$$

↓  
Direction  
(Screw Rule)

Unit

$$\vec{L} = \vec{r} \times \vec{p} = m \text{ kg m/s} = \text{kg } \frac{\text{m}^2}{\text{s}}$$

Dimension

$$\vec{L} = \vec{r} \times \vec{p} = L \times M L T^{-1} = M L^2 T^{-1}$$



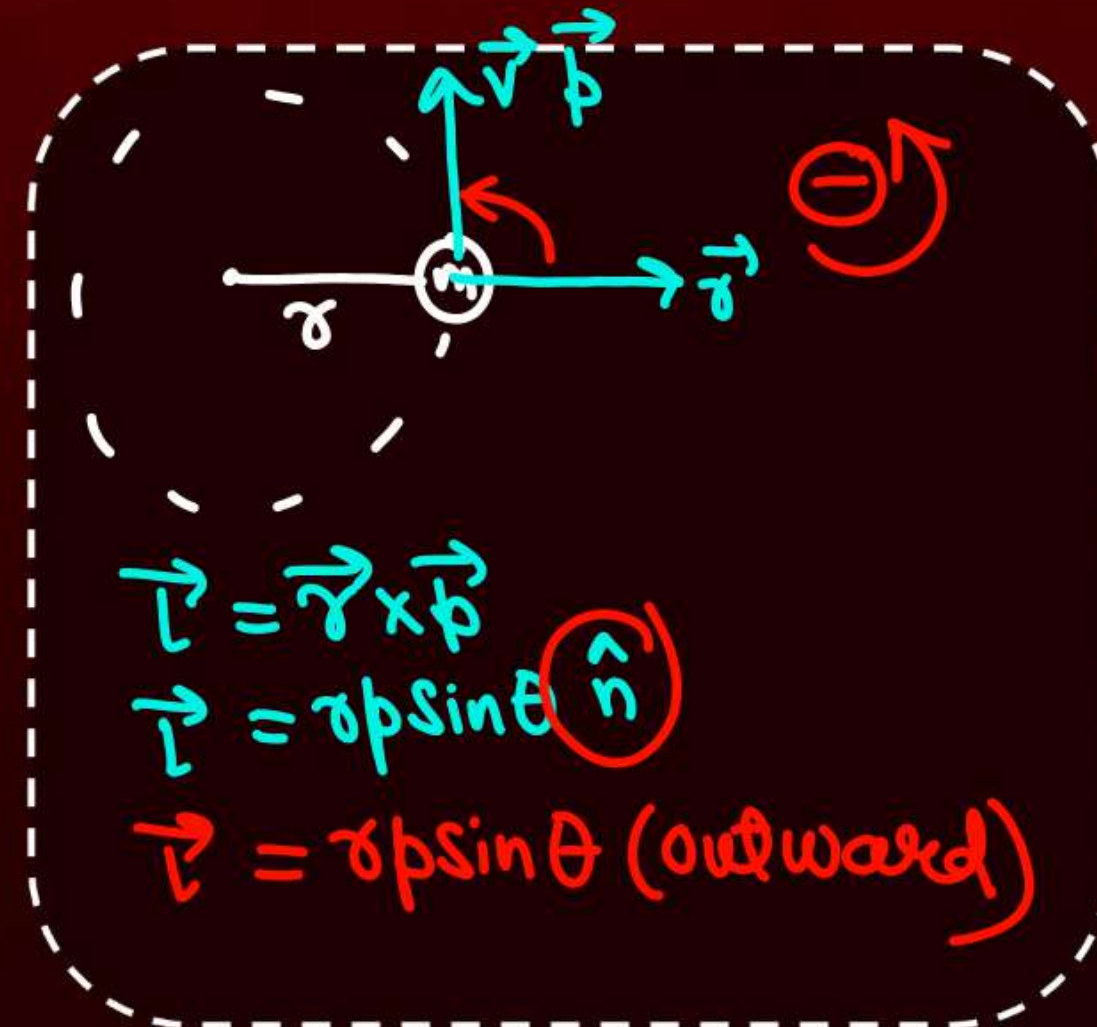
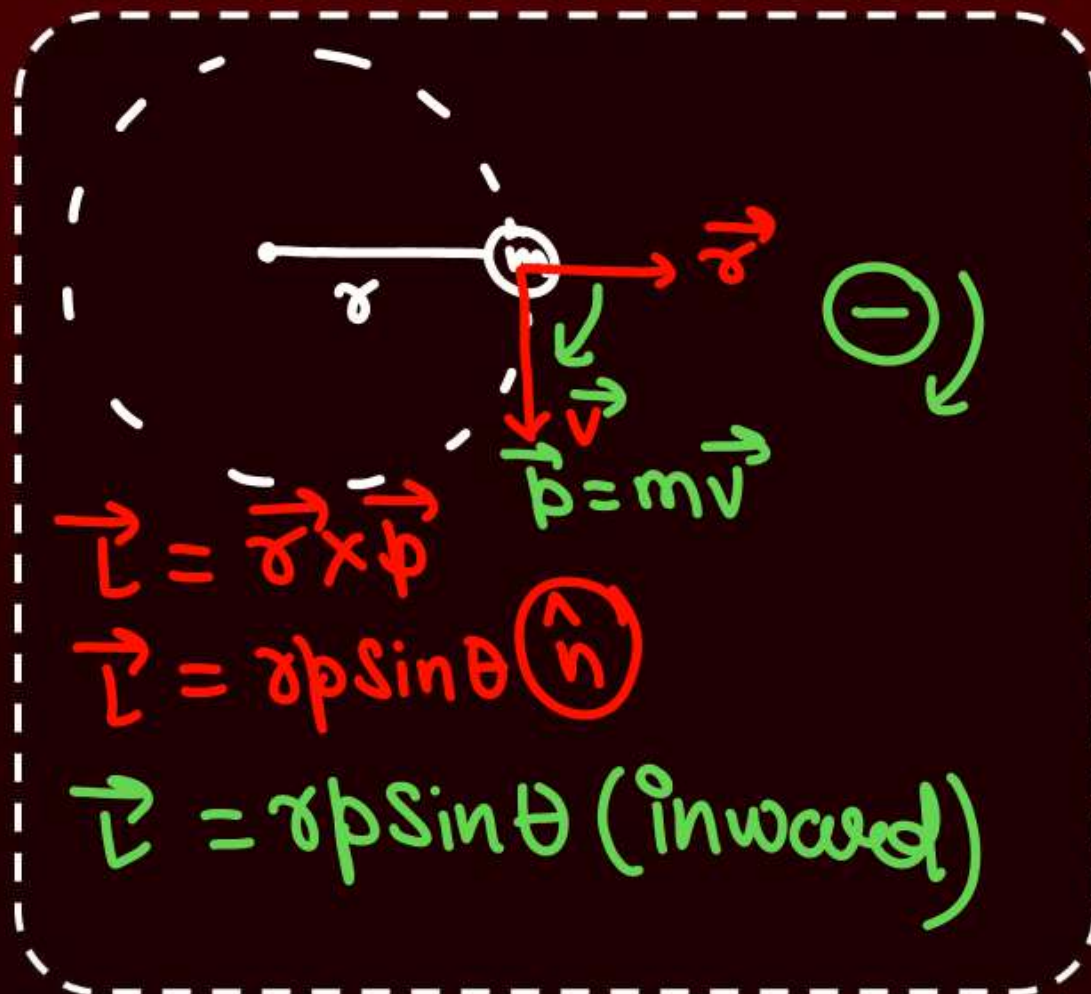


## Direction Of Angular Momentum

- ★ The direction of  $\vec{L}$  is give by Screw Rule. ✓  
 $\vec{L} = \vec{r} \times \vec{p}$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = \vec{r} \times \vec{F}$$



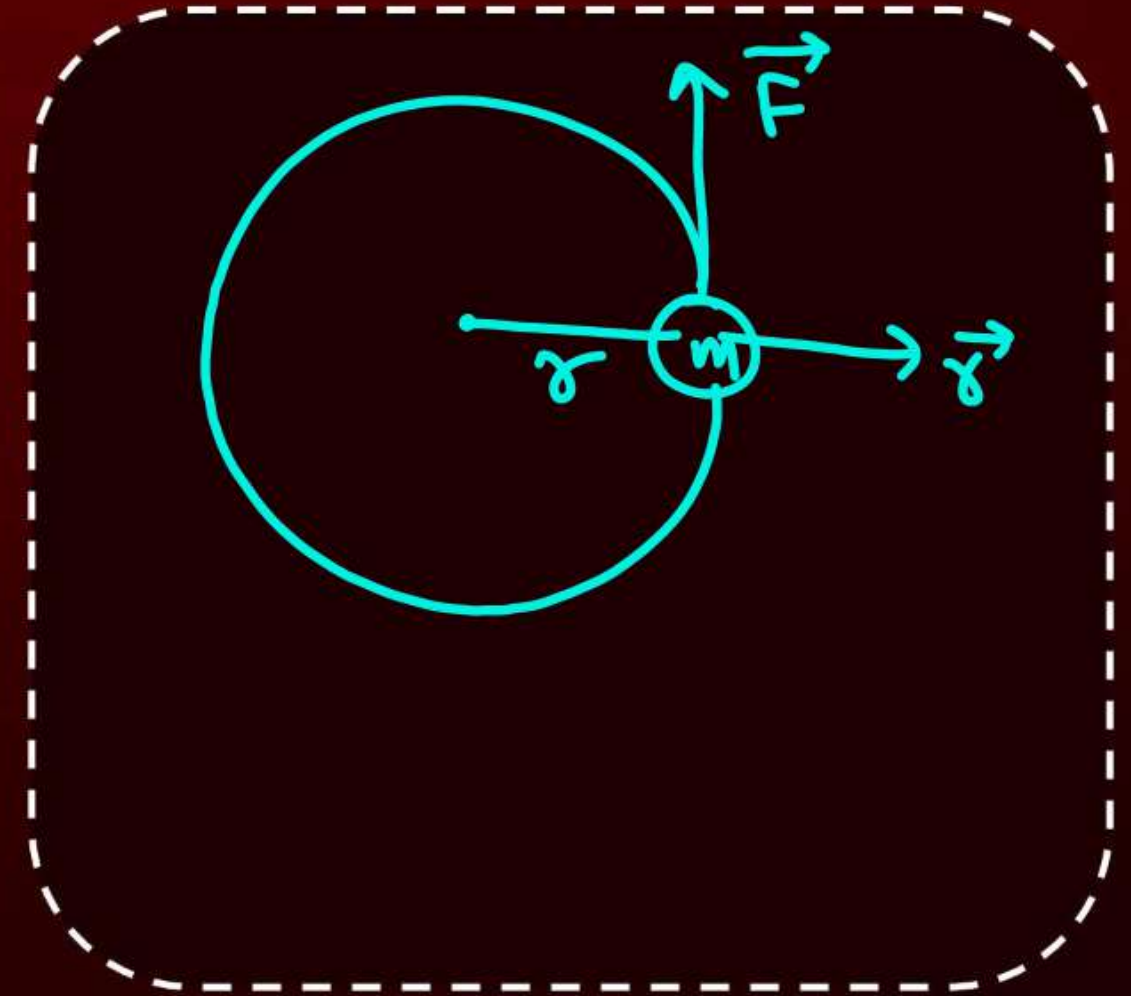


## Torque ( $\vec{\tau}$ )

- ✦ The turning effect of the force about an axis of rotation is called "**Torque**".
- ✦ Torque is the cross product of  $\vec{r}$  and  $\vec{F}$

$$\begin{aligned}\vec{\tau} &= \vec{r} \times \vec{F} \\ \tau &= r F \sin\theta \hat{n}\end{aligned}$$

Direction  
↓  
Screw Rule







# Torque ( $\vec{\tau}$ )

Unit

2024

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = m\vec{v}$$

$$\vec{\tau} = Nm$$

Shagufta

WORK | Energy

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = L m L T^{-2}$$

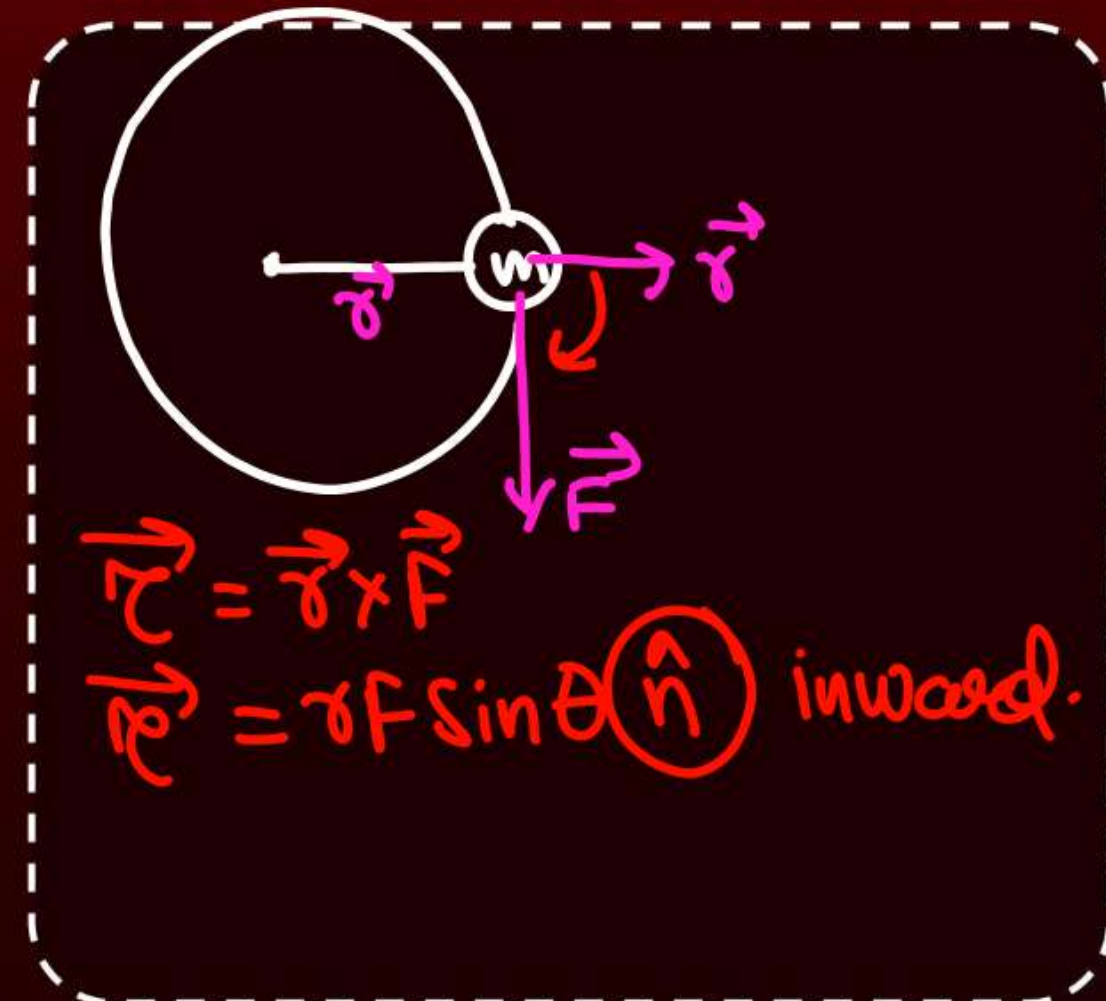
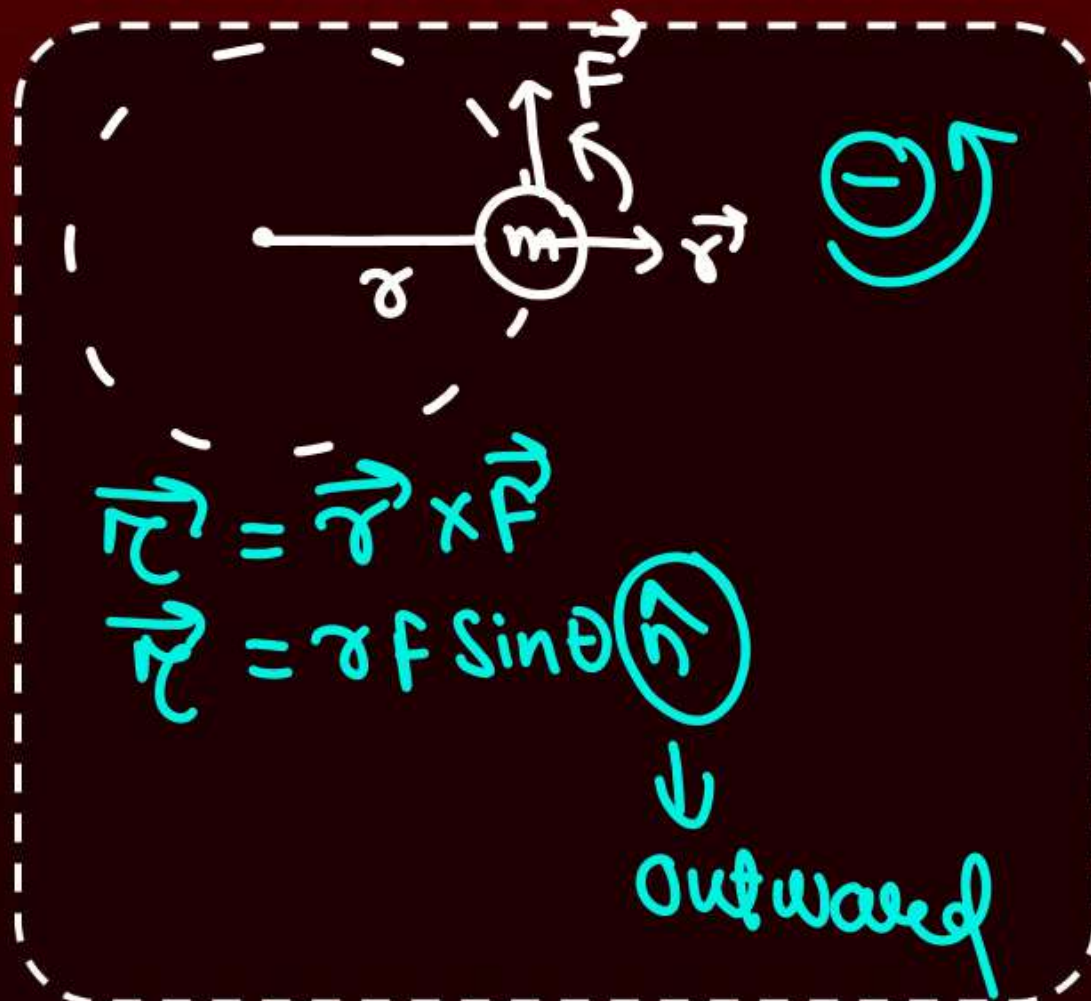
$$\vec{\tau} = m L^2 T^{-2}$$

Dimension



## Direction Of Torque

- ✦ The direction of  $\vec{\tau}$  is given by Screw Rule.







# Torque

→ 3M → 2021  
2022  
2023  
2016

- ✦ Deriving the relationship between Torque and angular momentum

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

"Newton's second law of motion"

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\underline{\vec{r} \times \vec{F}} = \vec{r} \times \frac{d\vec{p}}{dt}$$

$$\vec{\tau} = \frac{d\vec{r} \times \vec{p}}{dt}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

**"Torque is rate of change of angular momentum"**



## Rotational Equilibrium of the Body

- ★ When net torque to zero the Body is said to be in Rotational Equilibrium

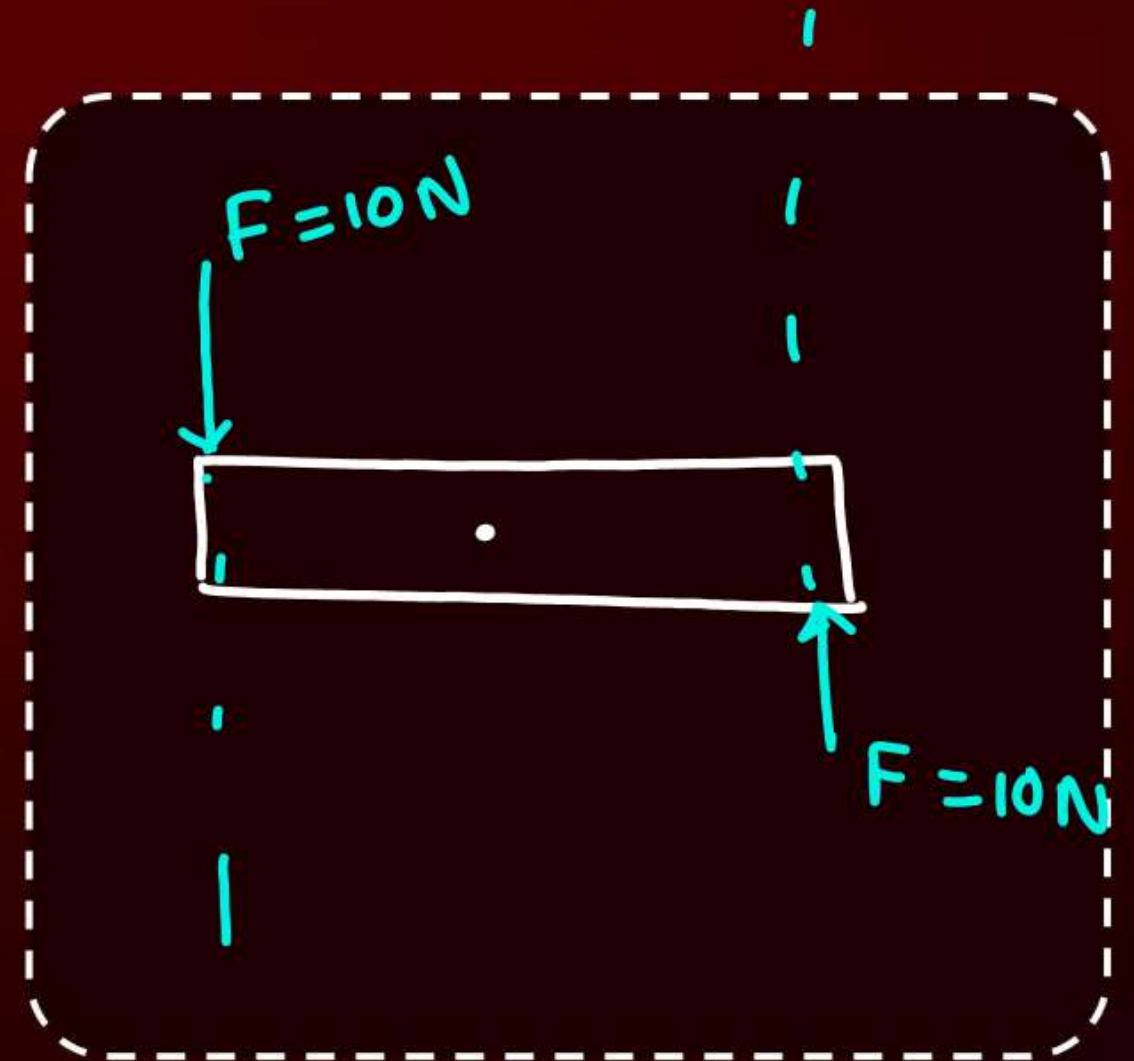
$$\underline{\tau_{\text{net}} = 0}$$





## Couple

- ✦ A couple or a force couple refers to two parallel forces that are equal in magnitude, opposite in sense and do not share a line of action.
- ✦ It causes the rotation. ✓





# Moment Of Inertia

- ★ The moment of inertia is a measure of how resistant an object is to changes in its rotational motion.

$$I \propto m \text{ — ①}$$

$$I \propto R^2 \text{ — ②}$$

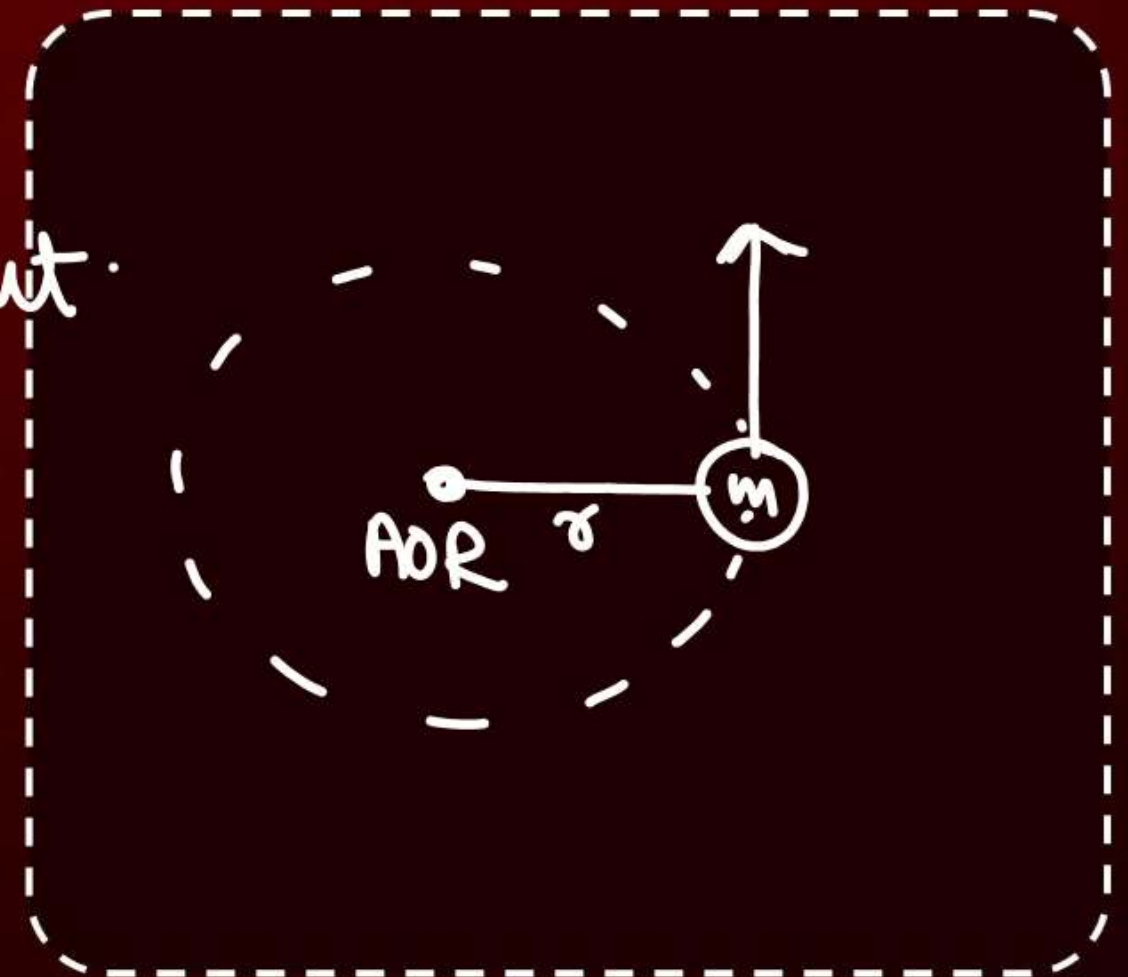
$$I \propto mR^2$$

$$I = \underbrace{k}_{\text{constant}} m R^2$$

$$I = 4 m R^2$$

$$I = m R^2$$

moment of Inertia is Inertia  
in Rotation ✓







# Moment Of Inertia

**Unit**

$$I = \underline{m} \underline{R^2}$$

$$I = \text{kg m}^2$$

**Dimension**

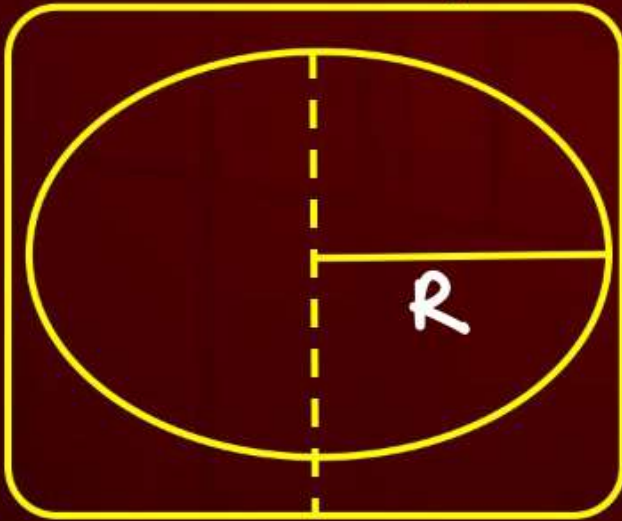
$$I = \underline{m} \underline{R^2}$$

$$I = \underline{M} \underline{L^2}$$



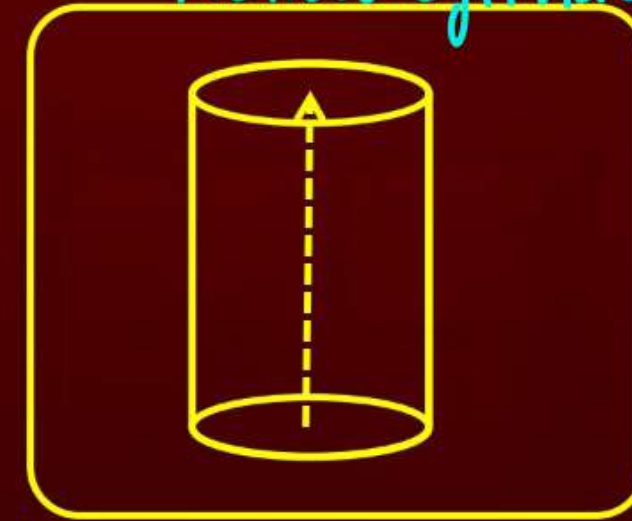
# Moment of Inertia Of Different Bodies

Ring



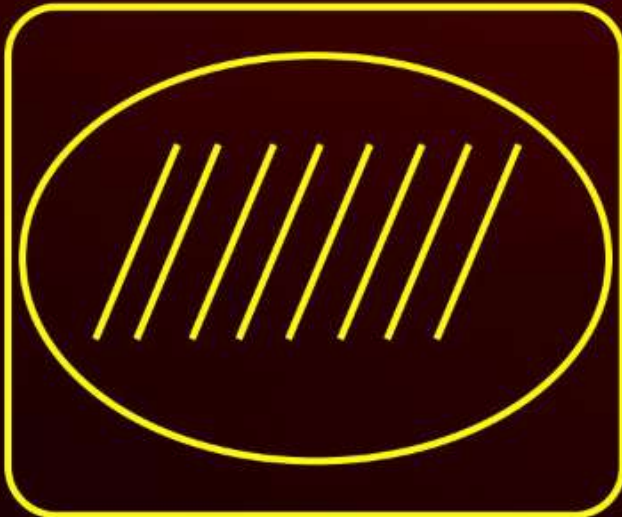
$$I = mR^2$$

Hollow Cylinder



$$I = mR^2$$

disc

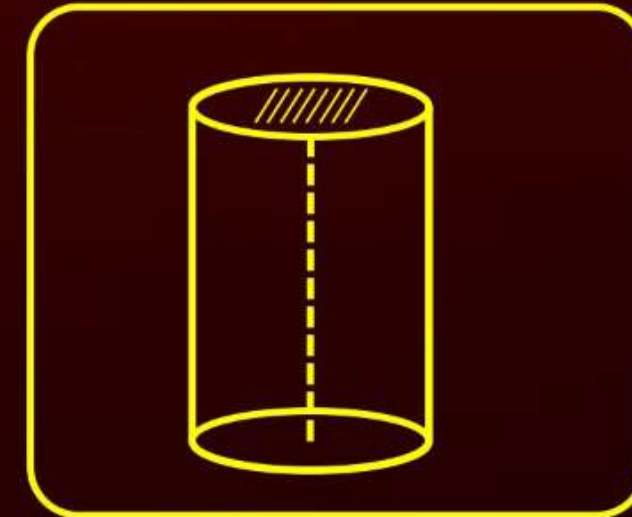


$$I = \frac{1}{2}mR^2$$

$$I = \frac{mK^2}{2}$$



Solid cylinder



$$I = \frac{mR^2}{2}$$



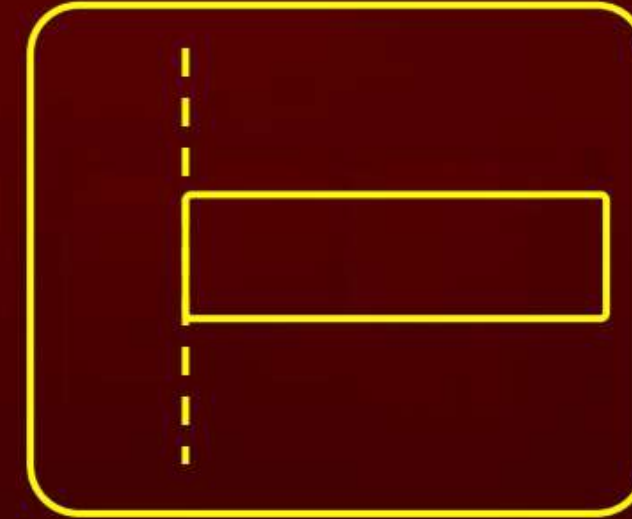


# Moment of Inertia Of Different Bodies

Rod

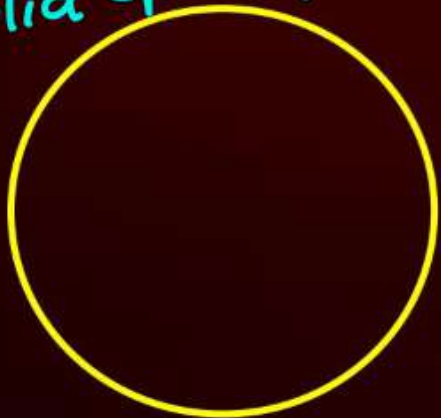


$$I = \frac{mL^2}{12}$$



$$I = \frac{mL^2}{3}$$

Solid sphere



$$I = \frac{2}{5}mR^2$$

Hollow Sphere



$$I = \frac{2}{3}mR^2$$



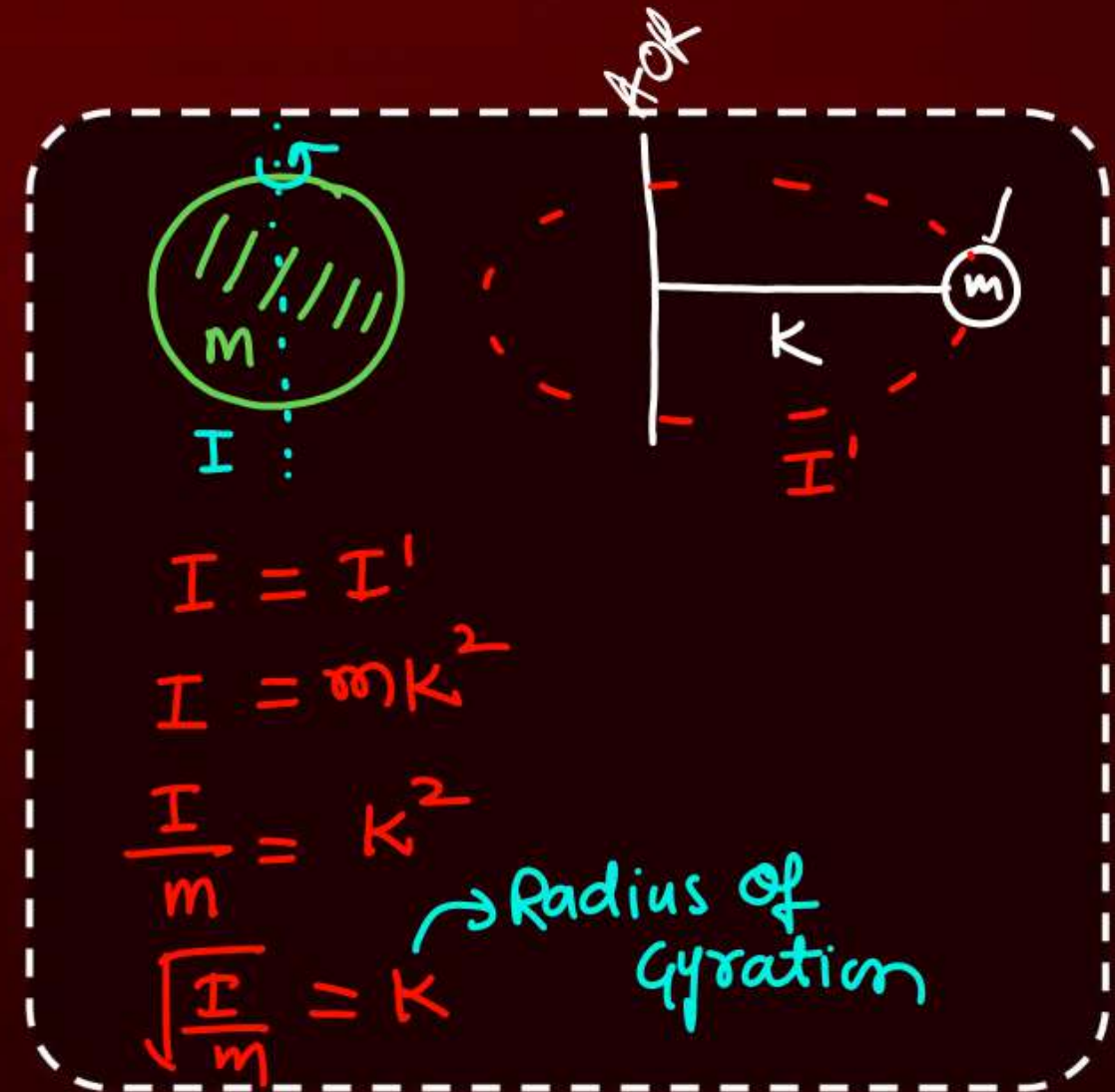


## Radius of Gyration

- ★ Radius of Gyration is the distance of  $m$  from Axis of Rotation so that its Initial moment of Inertia ( $I$ ) is equal to New moment of Inertia ( $I'$ ) when Rotated

2021

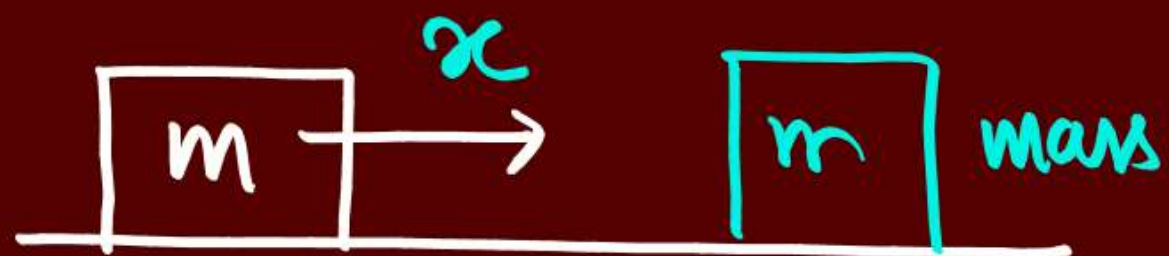
Unit  $K \rightarrow$  distance  
 $K = \text{metre}$





## Translation Motion

When Body moves in straight line



$$v = \frac{x}{t}$$

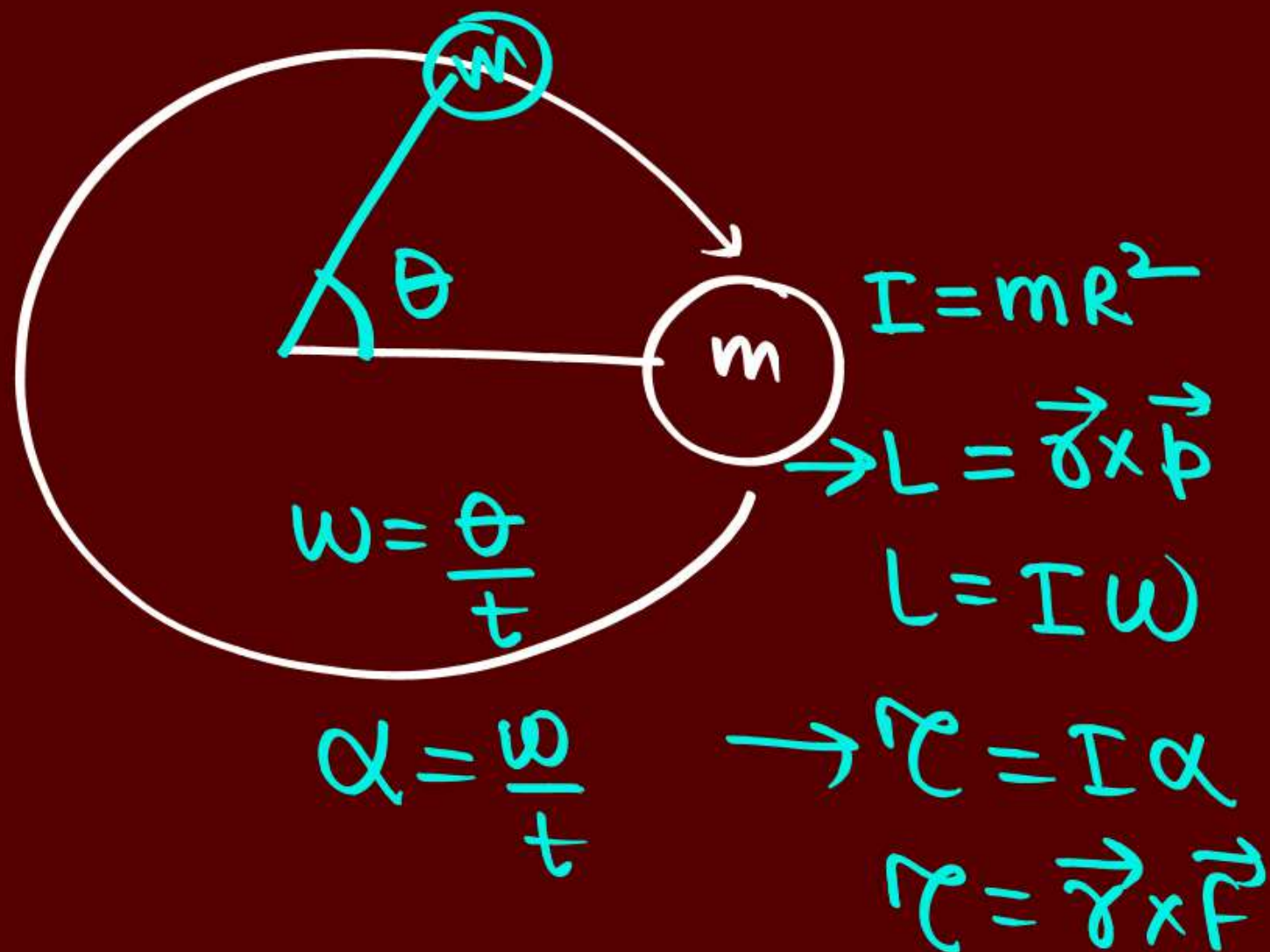
$$p = mv$$

$$a = \frac{v}{t}$$

$$f = ma$$

## Rotational Motion

When Body moves around the circular path about an axis of Rotation







## Comparison Between Rotational And Translation Motion

| S.No | Translational Motion                | Rotational Motion about a fixed axis                                                 |
|------|-------------------------------------|--------------------------------------------------------------------------------------|
| 1    | <u>Displacement X</u>               | <u>Angular Displacement, <math>\theta</math></u>                                     |
| 2    | <u>Time, t</u>                      | <u>Time, t</u>                                                                       |
| 3    | Velocity, $v = \frac{dx}{dt}$ $u/v$ | $\omega_0/\omega$ Angular velocity, $\omega = \frac{d\theta}{dt} = \frac{\theta}{t}$ |
| 4    | Acceleration, $a = \frac{dv}{dt}$   | Angular Acceleration, $\alpha = \frac{d\omega}{dt}$                                  |
| 5    | Mass, m                             | Moment of Inertia, $I = mr^2$                                                        |





## Comparison Between Rotational And Translation Motion

| S.No | Translational Motion                      | Rotational Motion about a fixed axis                            |
|------|-------------------------------------------|-----------------------------------------------------------------|
| 6    | Mass, $m$ $F = ma$                        | Torque $(\tau) = I\alpha$ $\vec{\tau} = \vec{r} \times \vec{F}$ |
| 7    | $P = mv$ (momentum)                       | Angular Momentum, $L = I\omega$ $L = \vec{r} \times \vec{p}$    |
| 8    | $\vec{F} = \frac{d\vec{p}}{dt}$           | $\vec{\tau} = \frac{d\vec{L}}{dt}$                              |
| 9    | Work done, $w = Fs$                       | Work done, $w = \tau\theta$                                     |
| 10   | Kinetic Energy, K. E. = $\frac{1}{2}mv^2$ | Kinetic Energy, K. E. = $\frac{1}{2}I\omega^2$ (mcq 2025)       |
| 11   | Power, $P = Fv$                           | Power, $P = \tau\omega$                                         |





## Kinematic Equation of Rotational Motion

1

$$v = u + at$$

$$\omega = \omega_0 + \alpha t$$

2

$$s = ut + \frac{1}{2}at^2$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

3

$$v^2 = u^2 + 2as$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$





# Law of Conservation of Angular Momentum

Angular

2025  
2024  
2023

2022  
2021



- ★ When external torque is zero then initial angular momentum is equal to final angular momentum.

$$\tau = \frac{d\vec{L}}{dt}$$

$$0 = \frac{d\vec{L}}{dt}$$

$$0 \times dt = d\vec{L}$$

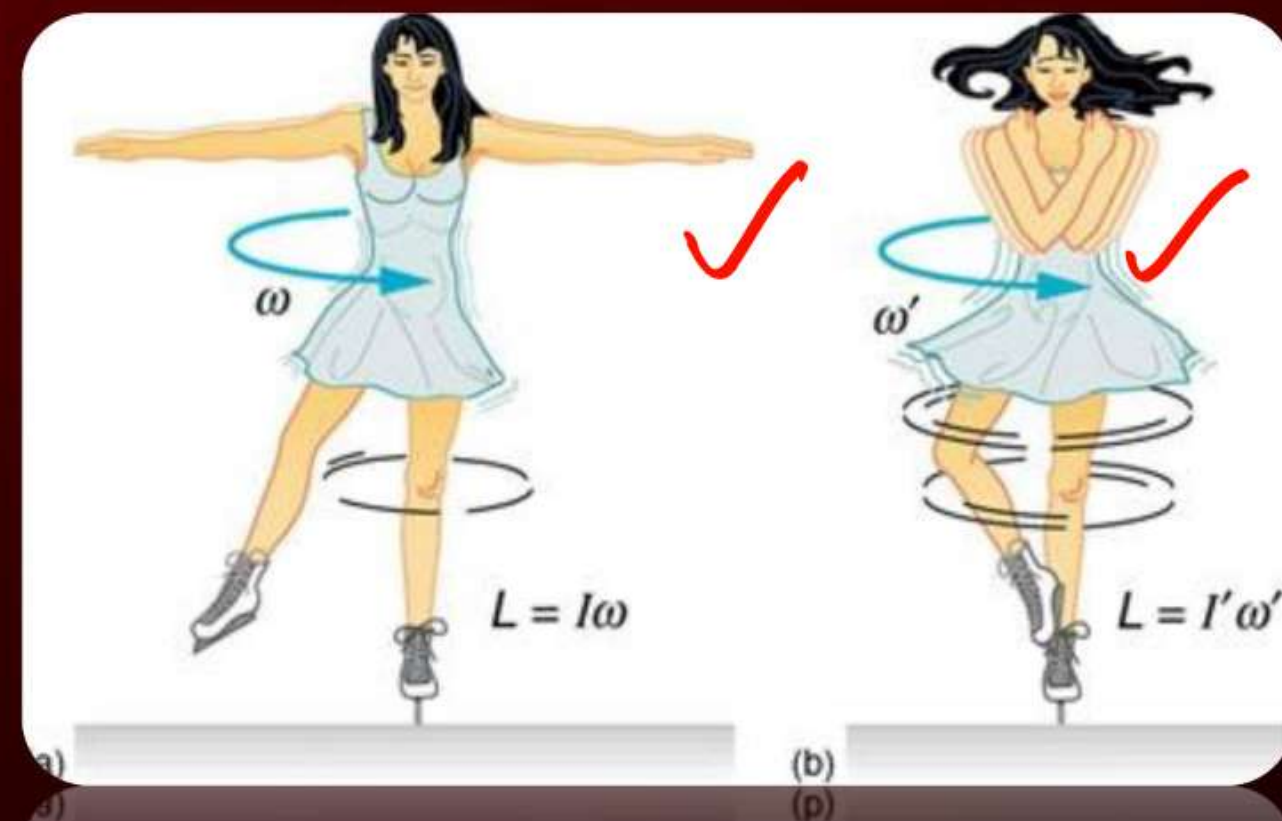
$$0 = d\vec{L}$$

$$d\vec{L} = 0$$

$$\Delta\vec{L} = 0$$

$$L_f - L_i = 0$$

$$L_f = L_i$$







## Law of Conservation of Momentum

$$L = I \omega$$



$$L_i = L_f$$

★  $I_i \omega_i = I_f \omega_f$

↓↑  $I_i = I_f \omega_f$

↓↑  $\omega_i$

$I \propto \frac{1}{\omega}$

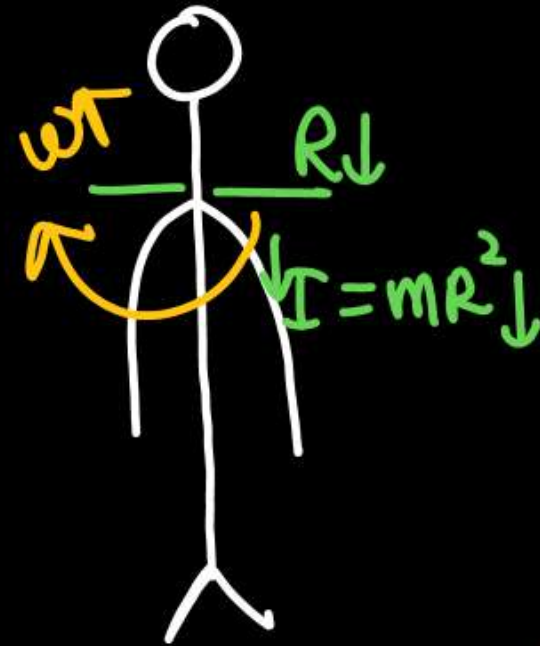
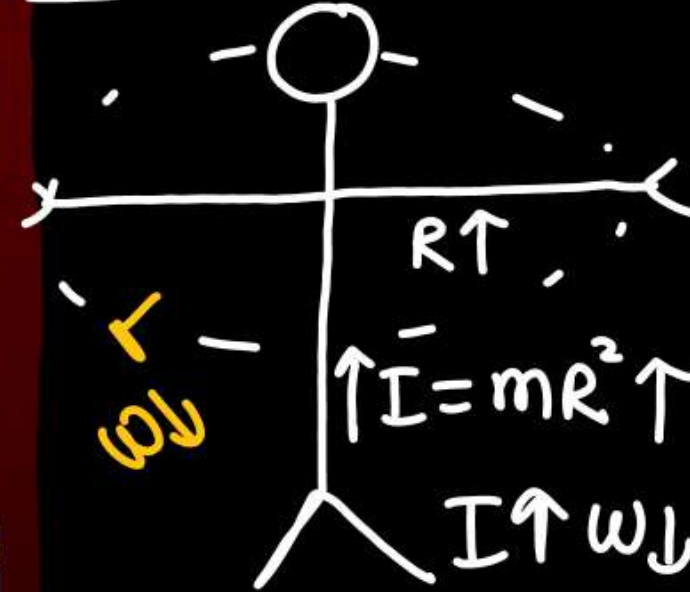
### CASE 1

✓  $R \uparrow I \uparrow \omega \downarrow$

### CASE 2

✓  $R \downarrow I \downarrow \omega \uparrow$

(Application)  
case-①







## Question [2025]

H/W ✓

A torque of 20 N-m is applied on a wheel initially at rest. Calculate angular momentum of the wheel after 3 seconds.

$$R_s \rightarrow 10 / \text{---}$$





## Question [2023]

**State and explain the principle of conservation of angular momentum.**





**Thank  
You**